

Diploma in Information Technology

HANDBOOK

DIT 2166 IT PROJECT

Introduction

The purpose of this subject is to check the skills learned by the students in the course of their study for the diploma. This subject requires the students to analyse, design and implement a simple system. The project provides the student hands-on experience of the project development life cycle to ensure the development of quality deliverables and documentation for the user. It gives them the opportunity to develop the ability to analyse the requirements for a project, and to evaluate various design options and to recommend an appropriate design that fulfils the users' needs. Students are also given the opportunity to develop the ability to plan, prepare and deliver written and oral presentations of a professional standard to a user audience within the stipulated duration.

Learning Outcomes

Upon completion of this subject, students should be able to:

- Apply general approach to start on a project of interest, the need for proper documentation and reporting, and professional presentation of plans and proposals writing.
- Analyse user needs in the selection, creation, evaluation and administration of computer-based system.
- Select suitable tools/methods for project development by using appropriate specifications/models
- Develop a system or theorem based on the selected specifications or models
- Evaluate process involved in using computers to solve practical problems.
- Present the project outcomes through a written report

Project Topic Selection

The selection of an appropriate project topic constitutes an important task undertaken by students prior to the commencement of the project work. A project may be theoretical, practical in nature or a combination of both. Generally, careful consideration should be given to such critical factors as the relevance of the topic to the programme undertaken, complexity of the problem, student's own interests and career aspirations, and the availability of a supervisor.

A project topic may come from:

1. The published list of project topics and their description.
2. Members of the academic staff
3. A student's own idea

Choosing from the published list:

The topic list will include a brief description of the project and associated supervisor. If a student decides to choose a topic from the list, they will have to fill in a **Supervision Agreement Form** (see **Appendix B, Form 1**) indicating their preferences of which they will be assigned a topic based on that and obtaining respective supervisor signature.

Choosing from the members of the academic staff or student's own idea:

If a student decides that they would like to work on their own topic (this can be from members of the academic staff, student's own idea), they will have to select a supervisor and discuss the topic with them. It is the supervisor's duty to ensure the topic is up to standard. Student has to fill in a **Project Detail Form** (see **Appendix B, Form 1**) stating "**OWN**" under project code and obtaining selected supervisor signature.

Project Supervision

Students must be supervised by a full-time DIIT academic staff or any with IT related discipline academic staff. A project undertaken without the supervision by such a supervisor will NOT be considered for assessment.

The supervisor advises students and points to the strategic directions of the work involved. Students are expected to work out the details of the project and carry on with the work unaided. The supervisor assists and monitors the progress of the work and is responsible for the assessment. The supervisor is also concerned with such factors as student's general approach and enthusiasm, ability to identify and discuss problems, recognition of deviations from the work plan and corrective actions taken, general practical and theoretical ability, etc. It is the **responsibility of the student** to meet and report to the supervisor regularly, e.g. on a weekly basis for an hour.

The major discussion points of project meetings should be recorded in the **Project Progress Form** (see **Appendix B, Form 2**). This form is to be kept by student after obtained supervisor signature and attached as Appendix in the Final Report upon the completion of the project.

The Project

The Project Proposal

The work on the project begins with the preparation and submission of a project proposal. The Flowchart of Project may be found in **Appendix A**. No more than ten A4 pages long, the student must follow the required template for the title page (see **Appendix C**). No other texts or illustrations, apart from those shown on the template, should be used on the title page. With the exception of the Sunway College logo, there must not be any use of colour on the title page. The **Project Proposal** should be submitted to the Blackboard e-Learn.

The sub-topics that are required in the **Project Proposal** include

- Introduction
- Problem Statement (with initial literature review and references)
- Objectives
- Scope of Work
- Methodology
- Proposed Outcome and Timeline
- Risk Management
- References

It is possible to include more sub-topics but there should not be fewer than what has been prescribed. The sample descriptions of the contents / rubrics of each of these sub-topics may be found in **Appendix D** and **Appendix G**.

In addition, the student is reminded that he/she should always consult and seek advice from the project supervisor regarding the specific contents and overall requirements of the proposal which may vary depending upon the project topic chosen.

The student is reminded that proper referencing is required throughout the **Project Proposal**. The **Project Proposal** is worth 100 marks. This contributes 100% of the final Project mark.

Progress Reports

Five Progress Reports are required to indicate the status of the Project. These reports should follow the format of the Final Report (**Appendix C**, and **Appendix D**) as much as progress will allow. Each Progress Report is worth 2 marks. This contributes 10% of the final Project mark.

The Final Report

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This represents the formal documentation of the Project. The required template for the title page may be found in **Appendix C**. No other texts or illustrations, apart from those shown on the template, should be used on the title page. With the exception of the Sunway College logo, there must not be any use of colour on the title page. The dimensions of the page margins are identical to those of the main body of the **Final Report**.

The **Final Report** should be submitted to the Project coordinator. The administrative procedure for **Submission of Final Report** is in **Appendix F**.

The sub-topics that are required in the **Final Report** include

- Abstract
- Table of Contents
- Introduction
- Literature Review
- Methodology
- Work Done
- Discussion of Results
- Conclusions
- References

It is possible to include more sub-topics but there should not be fewer than what has been prescribed. The sample descriptions of the contents / rubrics of each of these sub-topics may be found in **Appendix D**.

In addition, the student is reminded that he/she should always consult and seek advice from the project supervisor regarding the specific contents and overall requirements of the proposal which may vary depending upon the project topic chosen.

The **Final Report** must clearly show the personal contribution of the student to the work undertaken.

Collated works from existing published sources will not normally constitute a viable project.

The student is advised to upload the draft of the **Final Report** to the Turnitin System (<http://turnitin.com/static/index.php>). The Turnitin Report must be generated for viewing by the Project coordinator.

The student is reminded that proper referencing is required throughout the **Final Report**.

The **Final Report** is worth 100 marks. This contributes 30% of the final Project mark.

Poster Presentation

Student required to prepare presentation poster for their project and present it on poster presentation session. The presentation will be chaired by the supervisor and Co-Examiner, should last no more than 45 minutes, usually 20-30 minutes for the Oral Presentation and the demonstration of the Working System. The balance would be for the "Question and Answer" session. However, the actual organization of the **Final Presentation** may depend upon the specific project and this is at the discretion of the Supervisor and the Co-Examiner. There may also be invited guests attending the **Poster Presentation**.

The student is expected to demonstrate his/her knowledge of the work undertaken and to defend the Final Report. He/she is also expected to demonstrate the required level of understanding and originality of the work presented and to clarify any points of misunderstanding.

The sample descriptions of the requirements / rubrics of the Final Presentation can be found in **Appendix D** and **Appendix G**.

The Working System

This represents the product / deliverable / system solution that is the result of

- Applying the **Methodology** of choice, and
- The **Work Done**

as described in the **Final Report**.

It is imperative that extensive testing be carried out to ensure that the system fulfils the **Objectives**. This would ensure a successful demonstration during the **Presentation**.

The areas of concern include

- Application of Methodology
- Development Tools
- Demonstration
 - Validation Test Sets
 - Overall Impression

The **Working System** is worth 100 marks. This contributes 50% of the final project mark.

The Oral Presentation

This concerns the presentation and communication skills of the student.

The Oral Presentation is worth 10 marks. This contributes 10% of the final Project mark.

Marking Scheme

Assessment Methods and Types	Distribution of Marks
Proposal (100 marks) – IT Project 1 • Introduction (10 marks) • Problem Statement (with initial literature and references) (20 marks) • Objectives (15 marks) • Scope of Work (15 marks) • Methodology (20 marks) • Projected Outcome and Timeline (with Gantt Chart) (10 marks) • Risk Management (10 marks)	100%
Progress Report (5 marks) – IT Project 2 • Format to follow given template. • 2 reports (5 marks each)	10%
Final Report (100 marks) – IT Project 2 • Abstract (10 marks) • Table of Contents (5 marks) • Introduction (10 marks) • Literature Review and References (20 marks) • Methodology (15 marks) • Work Done (15 marks) • Discussion of Results (15 marks) • Conclusion (10 marks)	30%
Final Presentation – IT Project 2 Working System (100 marks) • Application of Methodology (20 marks) • Development Tools (30 marks) • Demonstration - Validation Test Sets (30 marks) - Overall Impression (20 marks) Oral Presentation (10 marks)	60% (a: 50%) (b: 10%)
Final Total	100%

NOTE: Plagiarism in a project is a very serious offence and will nullify any marks awarded.

References

There are many systems for the citation of references. For this project report, students are encouraged to use the **Harvard System or IEEE System**.

In the Harvard System, the author's surname and year of publication are cited in the text of your report. The full details of the reference are included in a reference list at the end of the reference list. The list is listed in alphabetical order.

For details on guide to the Harvard System of referencing, student can access online "Guide to the Harvard Style of Referencing" from Anglia Ruskin University:

<http://libweb.anglia.ac.uk/referencing/harvard.htm> [Accessed 4 September 2014].

In the IEEE System, it is numbering system. Each citation in the text is given a unique number written in square brackets, example: [3]. Each citation is numbered in the order in which it appears in the text of your report. If you need to cite reference more than once in the text, the number of its first appearance (its unique number) is used each time you cite it. The full details of the reference are included in a reference list at the end of the reference list.

The list is not listed in alphabetical order. It is a list numbered from [1] to [n], the number of each listing corresponding to the unique number that each source was assigned in the text.

For more details on guide to use IEEE referencing system, student can access online "IEEE Citation Reference" from IEEE organization:

<http://www.ieee.org/documents/ieeecitationref.pdf> [Accessed 4 September 2014]

Reading List:

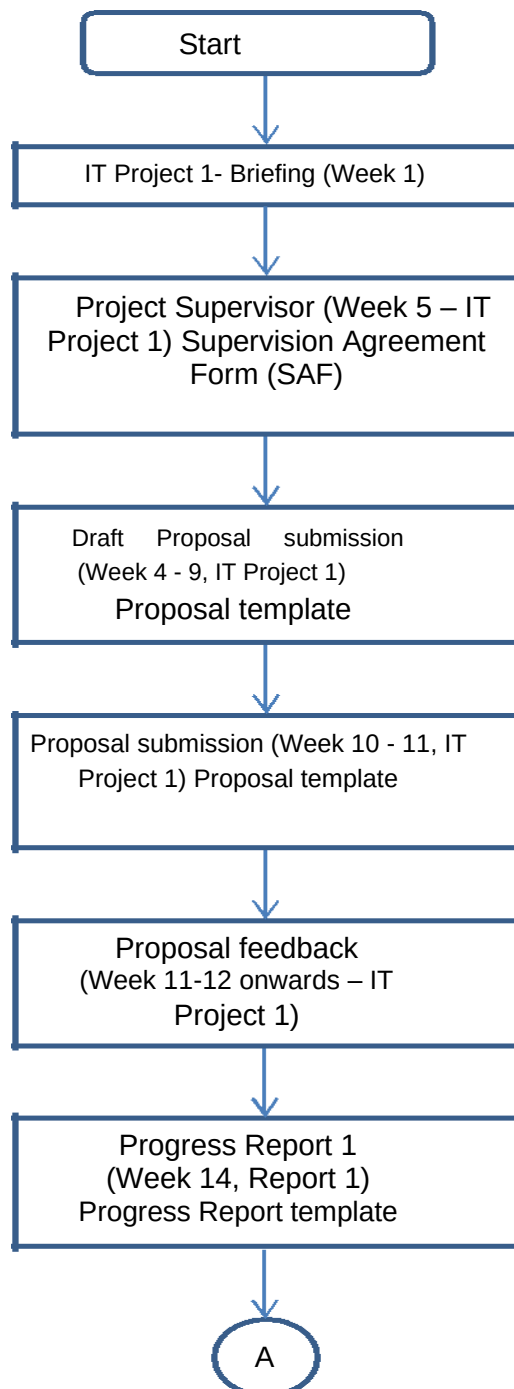
- Ringwood, J., 1999. Hints on Technical Report Writing. [Online] School of Engineering, Dublin City University. Available at: <http://odtl.dcu.ie/wp/1999/odtl-1999-03.html#s:refs> [Accessed 4 September 2014]
- Jain R., 2008. Project Guidelines. [Online] Washington University in Saint Louis. Available at: http://www.cse.wustl.edu/~jain/cse567-08/k_00prj.htm [Accessed 8 February 2010]



Appendix A

Flowchart of IT Project

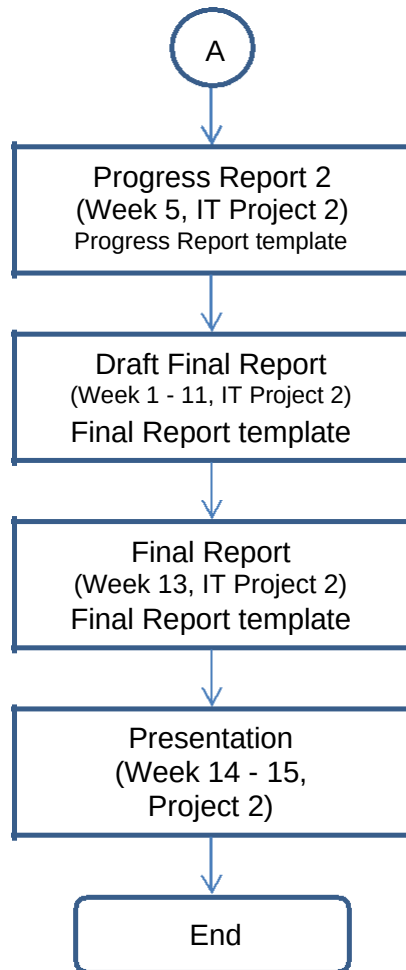
Timeline and Process:



****Remark:** Students are advised to discuss proposal in detail with supervisors. Students may submit as many drafts as required by their supervisors until their supervisors are agreeable to their proposal. Proposal template is available on e-Learn.



Timeline and Process:



****Remark:** Presentation slots will be announce on e-Learn. Each student is given 20-30 minutes presentation, and 15 minutes for Q&A.

Appendix B

Form 1 – Supervision Agreement Form

SUPERVISION AGREEMENT FORM

Instructions to student:

1. Please fill out your name and ID and send a copy to your supervisor's e-mail. You will have to request them to fill in details of the project.
2. Please ensure that all fields are filled EXCEPT for the fields "Moderator 1" and "Moderator 2".

Note to the student:

1. The marks shown are for work that has been completed with proper documentation and defensible during the viva. Failure to defend the work during the viva will and can result in no marks awarded for the other assessed components.

Form 2 – Project Progress Form

Date :

Week :



Updates from the previous meeting:

Items discussed this meeting:

Work for the coming meeting:

Supervisor's Signature

.....

Student's Signature

.....

Appendix C - Templates

Title page for IT Project Proposal



Sunway
Diploma
Studies¹

IT Project Proposal²

(This shall be your project title
2nd line of your project title
3rd line of your project title)³

by⁴

(This is your name
2nd line of your name - if applicable)⁵

Diploma in Information Technology⁶

Supervisor: (Change this to your supervisor's name)⁷

(Date of submission)⁸

* Margin for top, bottom, left and right: 2.54cm, centre alignment, font type: Arial Narrow

* Sunway College logo: Left-Top corner, size: 1.52 x 3.79 cm

* ALL items with TWO lines spacing in between, EXCEPT for last two items with TEN lines spacing and 1 line spacing.

¹ Font size 11 pt, Bold

² Font size 16 pt, Bold

³ Font size 14 pt, Bold

^{4, 5, 6, 7, 8} Font size 14 pt, normal

Title page for Final Report



**Sunway
Diploma
Studies¹**

IT Project²

**(This shall be your project title
2nd line of your project title
3rd line of your project title)³**

by⁴

(This is your name
2nd line of your name - if applicable)⁵

Diploma in Information Technology⁶

Supervisor: (Change this to your supervisor's name)⁷

(Date of submission)⁸

* Margin for top, bottom, left and right: 2.54cm, centre alignment, font type: Arial Narrow

* Sunway College logo: Left-Top corner, size: 1.52 x 3.79 cm

* ALL TWO lines spacing in between items, EXCEPT last two items with TEN lines spacing and 1 line spacing.

¹ Font size 11 pt, Bold

² Font size 16 pt, Bold

³ Font size 14 pt, Bold

^{4, 5, 6, 7, 8} Font size 14 pt, normal

Abstract

Project title : (Your title)
Student : (Your name)
Supervisor : (Your supervisor's name)

Abstract

(All fonts in your report should be font type of Arial Narrow with size 12pt. Font size for title/heading is 14pt and bold. Abstract should be about 200 words, and does not have a page number.)

Table of Contents (ToC)

Table of Contents

1 Introduction	1
1.1 Research/Work done	1
1.2 Contribution	2
2 Problem Statement and Literature Review	3
2.2 Problem Statement	3
2.3 Related works	3
3 Goals	4
4 Objectives	5
5 Scope of Work	6
6 Methodology	7
6.1 Requirement	7
6.1.1 Questionnaires	8
7 Projected outcomes with timeline	10
8 References	12

Appendix D

The Project Proposal

Introduction (10 marks)

This represents the overview of the proposed project. It introduces the following sub-topics:

- Problem statement (with initial literature review and references)
- Objectives
- Scope of work
- Methodology
- Proposed outcome and timeline

This section introduces the background to the work, starting from a broad discussion and then narrowing down to the **Project Proposal** question. It also touches on the structure of the **Project Proposal** according to the sub-topics. It sets the tone for the rest of the proposal and should be presented in the proper order. Proper referencing is required in this section.

Problem Statement (with initial literature review and references) (20 marks)

This section describes the problem at hand. This description must include relevant research of existing attempts at solving this problem. The problem could be any of the following:

- New or existing system that requires fresh development of an automated solution
- Survey research of state-of-the-art with a recommendation for value-added work in at least one significant aspect, e.g. new test measure, new toolset to facilitate learning
- Any fresh perspective of an existing problem that leads to a solution that meets changing commercial / technological needs

The literature review and references must conform to the required referencing style. Although the references are made primarily in the **Problem Statement**, it is important to note that the **"References"** section must be placed at the very end of the **Project Proposal**.

Objectives (15 marks)

The actual solution to the problem should be discussed here. This description should be measurable as much as possible. For example, if speed is to be improved, then there should be a comparison as to how much better the new approach is relative to the existing approach.

Scope of Work (15 marks)

The scope of work refers to the breadth (e.g. "How many departments would the system/service?") and depth (e.g.

"How detailed is the expected solution?"). This helps to put boundaries around the proposed project.

Methodology (20 marks)

This section describes the steps that are to be taken to develop the solution according to some existing approach (e.g. “Rapid Application Development”, on-line survey questionnaire) or in combination with a self-formulated technique (e.g. approach, materials, procedures, etc.). Key features of the methodology of choice should be discussed especially if these lead to the development of a robust solution. Include any limitations, exclusions and / or assumptions.

Projected Outcome and Timeline (with Gantt chart) (10 marks)

This describes the expected outcome of the project and should reflect that the project objectives have been successfully achieved. There should also be a work schedule, represented in a Gantt chart, which incorporates all task descriptions and milestones.

Risk Management (10 marks)

This section describes the potential risk students may face and the mitigation strategy for those risks.

References

All sources must conform to the required referencing style in **References**.

The Final Report

Abstract (10 marks)

This should enable a quick assessment of the subject matter of the report, the essentials of the work carried out and the main conclusions. The format of the abstract may be found in **Appendix C**. About 200 words long, the **Abstract** must be included in all reports. (**Note: Final Report submissions will not be accepted if the Abstract does not conform to the required format.**)

A copy of the **Abstract** will be indexed in an electronic database for future reference and research. It must, therefore, be able to stand alone as a document containing all the requisite information of the Project.

Table of Contents (5 marks)

A typical ToC should include the chapter, section and sub-section heading with page numbers, and clearly separated by successive dots and appropriate spacing. A sample ToC may be found in **Appendix C**. Note that the second level of division is usually sufficient.

Introduction (10 marks)

This represents the overview of the Report and the **Final Report**. It introduces the following sub-topics:

- Literature Review
- Methodology
- Work Done
- Discussion of Results
- Conclusions

This section introduces the background to the work, starting from a broad discussion and then narrowing down to the Project. It also touches on the structure of the **Final Report** according to the sub-topics. There must also be a clear statement of the objectives of the Project and how these objectives have been met. This will set the tone for the rest of the **Final Report** and should be presented in the proper order. Proper referencing is required in this section.

Literature Review and References (20 marks)

This requires an extensive discussion of related research, i.e. relevant research of existing / previous attempts at solving the problem introduced in the **Problem Statement** of the **Project Proposal**.

All sources of the information must be properly cited and they must conform to the required referencing style. Although the references are made primarily within the body of the **Final Report**, it is important to note that the "**References**" section must be placed at the very end of the **Final Report**.

Methodology (15 marks)

This section describes the steps that are to be taken to develop the solution according to some existing approach (e.g. “Rapid Application Development”, on-line survey questionnaire) or in combination with the student’s self-formulated technique (e.g. approach, materials, procedures, etc.). Key features of the methodology of choice should be discussed with the view of developing a robust solution. It is important to include limitations, exclusions and / or assumptions regarding the application of the methodology of choice.

Detailed extracts from original work need not be included here unless these are absolutely essential to the further development of the student’s original methodology. If not, such extracts may be included as an appendix. Proper referencing is required in this section.

Work Done (15 marks)

This section describes, in detail, the tasks and milestones stated in the **Project Proposal**, which were carried out in the development of the Project solution that fulfils the **Objectives**. Such activities might include analysis and design, the choice of development tools, pre- and post-processing of data, design of survey questions, survey data collection and analysis, software development, testing, troubleshooting, implementation, difficulties encountered, innovative solutions to overcome difficulties, etc. Proper referencing is required in this section.

Discussion of Results (15 marks)

The results arising from the work done is extensively analysed and discussed in relation to achieving the **Objectives**. Generally, one should include benchmark comparisons, typical / unusual results / behaviour patterns, accuracy and reliability issues, limitations, exclusions and / or assumptions of the methodology of choice, etc. Proper referencing is required in this section.

Conclusions (10 marks)

This summarizes the success of the project by emphasizing the impact of the results in achieving the **Objectives**. There must be no repetition from earlier sections, and no introduction of new material. Suggestions for future work should be included, if appropriate, by reflecting upon the **Discussion of Results** section.

References

All sources must conform to the required referencing style in **References**.

Appendix

Attach the project progress form here.

Final Presentation

A. Working System

- **Application of Methodology (20 marks)**
The system reflects a systematic and precise application of the **Methodology** discussed in the **Final Report**.
- **Development Tools (30 marks)**
The choice and combination of hardware and software are justified and these are reflected positively in the **Working System**.
- **Demonstration (50 marks)**
 - o **Validation test sets (30 marks)**
The choice of test sets validates all system features.
 - o **Overall impression (20 marks)**
The system operates with a professional look-and-feel. It reflects the **Work Done** described in the **Final Report**.

B. The Oral Presentation

▪ **Oral Presentation (10 marks)**

The points of interest include

- o Voice projection as this indicates the level of confidence in presenting as well as in answering questions
- o Clarity of explanations indicate the level of preparation for and the delivery of the information.
- o Command of language and use of appropriate terms

▪ **Final Presentation**

For project poster presentation, the minimum sub-topics are as below:

- o Introduction
- o Literature Review
- o Methodology
- o Final product (pictures & screenshots)
- o Conclusions

Appendix E

Additional Information for Mathematical Notation, Figures and Tables (Guidelines provided by the IEEE)

- **Mathematical Notation**

Preferably, use Microsoft Equation Editor. Avoid using Word fonts or symbols for in-line (within the flow of normal text) single variables with superscripts or subscripts to write an equation. Single (one-and-a-half line) spacing should be used to separate equations from text. All equations should be numbered consecutively with equation numbers in parentheses flush with the right margin (right aligned), as in (1). Equations should be punctuated when they part of a sentence.

- **Figures and Tables**

Figures, tables and all other illustrations used in the project should be sharp, noise free and of good contrast. All tables and illustrations should be centred on the page and separated from texts above and below by at least a single (one-and-a-half line) spacing. Tables and illustrations should have distinct sequences of numbering. Each may run continuously through the report, or the number of an illustration or table may incorporate the number of the chapter or section in which it is bound (e.g. the third illustration of chapter 4 may be numbered "Fig. 4.3"). Tables and illustrations should be placed nearer to where they are first mentioned in the text. There must not be any tables and illustrations in the report that have not been referred to at least once in the text.

Figure captions should be placed below the figures and table titles above the tables. A figure should be referred to in the text as "Fig." even at the beginning of a sentence. The word "Table" should not be abbreviated. To avoid any confusion in labelling graphs, axis labels should have words rather than symbols. Any units should be put in parentheses. For example, the "Output Voltage, V (V)" should be used not just "V (V)".

Appendix F

Submission of Final Report

Two copies of the report are required for the examination and will be retained by the department. Students are strongly advised to have a third copy available (softcopy) for reference during the Final Presentation and for their own use afterwards.

Late submissions will generally carry a penalty. However, exceptions may be entertained on a case by case basis. When a late submission is anticipated, students should obtain early approval for late submission by applying in writing using the Extenuating Circumstances Form (ECF) to the department. Failure to do so at the earliest opportunity could result in refusal to authorize a late submission. Documentary evidence to support the application (medical certificate, etc.) must be provided. **Final approval for late submission will be at the discretion of the Project Coordinator.**

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Appendix G

Assessment Rubrics

IT Project Proposal (100 Marks) - 100% of Total Marks					
	Level 1	Level 2	Level 3	Level 4	Marks Earned
	(0-49)	(50-59)	(60-69)	(70-100)	
Introduction (10 marks)	Weak overview of the proposed project. Only a few of the sub-topics (problem statement, objectives, scope of work, methodology, proposed outcome and timeline) are included.	Adequate overview of the proposed project. Most of the sub-topics (problem statement, objectives, scope of work, methodology, proposed outcome and timeline) are included although not all are present.	Good overview of the proposed project. The problem statement, objectives, scope of work, methodology, proposed outcome and timeline are written reasonably well and presented in proper order.	Exceptional overview of the proposed project. The problem statement, objectives, scope of work, methodology, proposed outcome and timeline are written in a clear precise manner and presented in proper order.	0
Problem statement (with initial literature review and references) (20 marks)	Poor description of the problem at hand, indicating little evidence of research. Numerous errors with insufficient number of sources cited, wrong and / or missing sources. Errors in the required referencing style.	Adequate description of the problem at hand, indicating research from a variety of sources. Quite a number of errors with insufficient number of sources cited. Errors in the required referencing style.	Good description of the problem at hand, indicating research from a variety of sources. A few errors with most sources properly cited. The occasional error in the required referencing style.	Excellent description of the problem at hand, indicating thorough and extensive research from a wide variety of sources. Error-free with all sources properly cited. Conforms to the required referencing style.	0

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Objectives (15 marks)	Poor description of the proposed resolution of the problem at hand. Numerous errors and written poorly.	Adequate description of the proposed resolution of the problem at hand. Quite a number of errors and written in a disjointed / disorganized manner.	Good description of the proposed resolution of the problem at hand. A few errors and written reasonably well.	Excellent description of the proposed resolution of the problem at hand. Error-free and written in a clear precise manner.	0
Scope of Work (15 marks)	Poor description of the breadth and depth of work that is to be carried out. Numerous errors and written poorly.	Adequate description of the breadth and depth of work that is to be carried out. Quite a number of errors and written in a disjointed / disorganized manner.	Good description of the breadth and depth of work that is to be carried out. A few errors and written reasonably well.	Excellent description of the breadth and depth of work that is to be carried out. Error-free and written in a clear precise manner.	0
Methodology (20 marks)	Weak justification of the specific choice of methodology, with brief mention of the approach required for deployment.	Adequate justification of the specific choice of methodology, with adequate mention of the approach required for deployment.	Good justification of the specific choice of methodology, briefly mentioning one, possibly two, key features that support its deployment.	Excellent justification of the specific choice of methodology, highlighting key features that support its deployment. Error-free and written in a clear precise manner.	0
Projected outcome and Timeline (with Gantt Chart) (10 marks)	Poor description of the expected outcome of the project. A disorganized work schedule.	Adequate description of the expected outcome of the project. A somewhat organized work schedule including some of the required task descriptions.	Good description of the expected outcome of the project. An organized work schedule including most of the required task descriptions.	Excellent description of the expected outcome of the project. A well-organized work schedule including all detailed task descriptions.	0

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Risk Management (10 marks)	Poor description of the expected risk and also the mitigation strategy.	Adequate description of the expected risk and the mitigation strategy.	Good description of the expected risk and the mitigation strategy.	Excellent description of the expected risk and also the mitigation strategy.	0
Final Total (out of 100 marks)					0

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Final Report (100 Marks) - 30% of Total Marks					
	Level 1 (0-49)	Level 2 (50-59)	Level 3 (60-69)	Level 4 (70-100)	Marks Awarded
Abstract (10 marks)	Weak explanation of the subject matter, the essential work to be conducted and the main conclusions. On the whole, the abstract provides a poor idea of the project.	Adequate explanation of the subject matter, the essential work to be conducted and the main conclusions. On the whole, the abstract provides a somewhat clear idea of the project.	Good explanation of the subject matter, the essential work to be conducted and the main conclusions. On the whole, a good abstract that sufficiently describes the project.	Exceptional explanation of the subject matter, the essential work to be conducted and proceeds seamlessly to the main conclusions. On the whole, an excellent abstract that compactly illustrates the entire project.	0.0
Table of Contents (5 marks)	A disorganized ToC. Barely conforms to the required format with numerous errors in spacing / section or page numbering / successive dots separating the numbers / spelling of headings / order of headings.	An adequately organized ToC c/w chapter, section and sub-section heading with page numbers. Somewhat conforms to the required format with several errors in spacing / section or page numbering / successive dots separating the numbers / spelling of headings / order of headings.	An organized ToC c/w chapter, section and sub-section heading with page numbers. Conforms to the required format with the odd error in spacing / section or page numbering / successive dots separating the numbers / spelling of headings.	A well-organized ToC c/w chapter, section and sub-section heading with page numbers, clearly separated by successive dots and appropriate spacing. Conforms to the required format with no errors.	0.0
Introduction (10 marks)	Weak overview of the project and the report. Only a few of the sub-topics (problem statement, project objectives, appropriate theories, work done, results and conclusions) are included.	Adequate overview of the project and the report. Most of the sub-topics (problem statement, project objectives, appropriate theories, work done, results and conclusions) are included although not all are present.	Good overview of the project and the report. The problem statement, project objectives, appropriate theories, work done, results and conclusions are written reasonably well and presented in proper order.	Exceptional overview of the project and the report. Succeeds in motivating the reader's interest to read on. The problem statement, project objectives, appropriate theories, work done, results and conclusions are written in a clear precise manner and presented in proper order.	0.0

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Literature Review and References (20 marks)	Weak review indicates little evidence of research. Numerous errors are present with insufficient number of sources cited, wrong and / or missing sources, as well as errors in the required referencing style.	Adequate review indicates research from a variety of sources. Quite a number of errors are present with insufficient number of sources cited, as well as errors in the required referencing style.	Good review indicates research from a variety of sources. A few errors are present with most sources properly cited, as well as the odd error in the required referencing style.	Exceptionally well written review indicates thorough and extensive research from a wide variety of sources. Overall, it is error-free with all sources properly cited and it conforms to the required referencing style.	0.0
Methodology (15 marks)	Weak justification of methodology, with a brief mention of the general approach required to resolve the problem. A disjointed / disorganized explanation indicates little understanding of the quoted choice. Little / Wrong / No referencing.	Adequate justification of methodology, with adequate mention of the general approach required to resolve the problem. A satisfactory explanation indicates some understanding of the quoted choice. A few references are included.	Good justification of methodology, with a brief mention of one, possibly two, characteristics that support the resolution of the problem. A proficient explanation indicates a good understanding of the quoted choice. Proper referencing is maintained throughout.	Excellent justification of methodology, highlighting particular characteristics that support the resolution of the problem. An extensive explanation indicates an excellent grasp of the quoted choice. Proper referencing is maintained throughout.	0.0
Work done (15 marks)	A disorganized description (would not include the majority of the analysis and design, development tools, pre- and post-processing of data, design of survey questions, etc.) of the development of the solution that does not completely fulfil the project objectives. Numerous errors exist. Little / Wrong / No referencing.	An adequately organized description (might include the analysis and design, development tools, pre- and post-processing of data, design of survey questions, etc.) of the development of the solution that fulfils some of the project objectives, with some milestones highlighted. Quite a number of errors exist. A few references are included.	A well organized description (includes the analysis and design, development tools, pre- and post-processing of data, design of survey questions, etc.) of the development of the solution that fulfils the project objectives, with key milestones highlighted and discussed. A few errors exist. Proper referencing is maintained throughout.	An exceptionally well organized, detailed and error-free description (includes the analysis and design, development tools, pre- and post-processing of data, design of survey questions, etc.) of the development of the solution that fulfils the project objectives, with key milestones highlighted and thoroughly discussed. Proper referencing is maintained throughout.	0.0

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Discussion of Results (15 marks)	The results are poorly analysed and discussed, not fulfilling most of the project objectives (might include one or two of the following: benchmark comparisons, typical / unusual results / behaviour patterns, accuracy and reliability issues, etc.). Numerous errors with few / wrong / no references included.	The results are adequately analysed and discussed in relation to fulfilling only some of the project objectives (might include benchmark comparisons, typical / unusual results / behaviour patterns, accuracy and reliability issues, etc.). Quite a number of errors with a few references included.	The results are well analysed and discussed in relation to fulfilling the project objectives (includes benchmark comparisons, typical / unusual results / behaviour patterns, accuracy and reliability issues, etc.). A few errors with proper referencing maintained throughout.	The results are extensively analysed and discussed in relation to fulfilling the project objectives (includes benchmark comparisons, typical / unusual results / behaviour patterns, accuracy and reliability issues, etc.). Error-free with proper referencing maintained throughout.	0.0
Conclusions (10 marks)	A weak summary that does not reflect positively on the fulfilment of the project objectives. Might not include mention of its shortcomings. Numerous errors with little or no idea for future work.	An adequate summary that discusses the success of the project in fulfilling some of the project objectives, as well as a brief mention of its shortcomings. Quite a number of errors with some vague ideas for future work.	A good summary that emphasizes the success of the project in fulfilling the project objectives. A few errors with some good ideas for future work.	An excellent summary that emphasizes the success of the project (with no repetition from earlier sub-sections) in fulfilling the project objectives. Error-free with researched ideas for future work.	0.0
Final Total (out of 100 marks)					0.0
Final Total (out of 40 marks)					0.0

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a. Working System (100 Marks) – 50% of Total Marks					
b. Oral Presentation (10 Marks) - 10% of Total Marks					
	Level 1	Level 2	Level 3	Level 4	Marks Earned
	(0-49)	(50-59)	(60-69)	(70-100)	
Working system (100 marks)					
Application of Methodology (20 marks)	The system reflects a weak application of the methodology discussed in the Final Report	The system reflects an adequate application of the methodology discussed in the Final Report.	The system reflects a good application of the methodology discussed in the Final Report.	The system reflects a systematic and precise application of the methodology discussed in the Final Report.	0.0
Development Tools (30 marks)	Weak justification of choice and combination of hardware and software, barely mentioning the general approach required to support system development.	Adequate justification of choice and combination of hardware and software, vaguely mentioning the general approach required to support system development.	Good justification of choice and combination of hardware and software, briefly mentioning one or, possibly two, characteristics that support system development.	Excellent justification of choice and combination of hardware and software, highlighting particular characteristics that support system development.	0.0
a) Demonstration - Validation test sets (30 marks)	Available test sets validate all system features with numerous errors.	Available test sets validate all system features with quite a number of errors.	Available test sets validate all system features with a few errors.	Available test sets completely validate all system features with no errors.	0.0
b) Demonstration - Overall impression (20 marks)	The system does not work at all / has numerous errors. It is a poor reflection of the work done described in the Final Report.	The system works adequately, with quite a number of errors. It is a reasonable reflection of the work done described in the Final Report.	The system works well, with the odd error. It is a good reflection of the work done described in the Final Report.	The system works exceptionally with a professional look-and-feel. It is an accurate and detailed reflection of the work done described in the Final Report.	0.0
Final Total (out of 100 marks)					0.0
Final Total (out of 50 marks)					0.0

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Oral Presentation (10 marks)	Great difficulty in communicating ideas, no confidence / inability in answering questions, poor voice projection, command of language, preparation and delivery.	Some difficulty in communicating ideas, some confidence in answering questions, adequate voice projection, command of language, preparation and delivery.	Communicates ideas reasonably well, confidence in answering questions, good voice projection, command of language, preparation and delivery.	Communicates ideas enthusiastically, great confidence in answering questions, excellent voice projection, command of language, preparation and delivery.	0.0
Final Total (out of 10 marks)					0.0